

Azhar Abdool

Software & Computer Engineer | Enterprise Software, Data & Intelligent Systems

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SUMMARY

Software Engineer at NCT Forestry and UCT Computer Science and Computer Engineering graduate. Builds enterprise applications, database workflows, integrations and GIS-enabled systems. Combines professional software delivery with practical foundations in machine learning, Foundry analytical workflows, embedded C, concurrency and networks. Currently completing part-time Honours study in Security & Network Engineering. A software-to-hardware engineering foundation with a growing focus on data-driven, intelligent systems.

TECHNICAL SKILLS

- **Software:** Java, JavaScript, Python, C, C++, C#, React Native, Oracle APEX, REST APIs, Firebase, Git, testing and debugging.
- **Data / Intelligent Systems:** SQL, Oracle, PL/SQL, pandas, NumPy, PyTorch, scikit-learn, classification, Q-learning, K-Means, PCA and model evaluation.
- **Enterprise Platforms:** Palantir Foundry, Ontology, AIP Logic, Gemini integration, data transformations; SAP S/4HANA and ABAP training.
- **Computer Engineering / Systems:** STM32, ADC, PWM, timers, DMA, Java concurrency, FCFS/SJF scheduling and virtual address translation.
- **Security / Networks:** TCP/UDP sockets, Wireshark, RBAC, input validation and application security; Azure security course training.

EXPERIENCE

Software Engineer | NCT Forestry

October 2025 - Present

- Build and maintain enterprise PIMS modules using Oracle APEX, PL/SQL and React Native.
- Develop database procedures, SQL reports, backend integrations and operational workflow logic.
- Implement role-based access control and input validation in enterprise application workflows.
- Develop GIS-enabled applications integrating Google Maps, GeoJSON and polygon validation with PL/SQL backend logic.
- Collaborate through Git workflows, debugging, testing and deployments; resolve geospatial rendering and integration issues.

SAP Graduate Programme | OpenSource Intelligent Solutions

October 2024 - March 2025

- Completed SAP Associate training and earned Business Process Integration with SAP S/4HANA certification.
- Participated in enterprise systems workshops and technical sessions, with ABAP on SAP BTP and ERP process learning records.

SELECTED ENGINEERING PROJECTS

Applied ML and Data Mining | Python, PyTorch, scikit-learn | 2024 / 2026

- Built a 784-128-64-10 MNIST classifier with minibatch training, evaluation and image prediction; original training log records 95.71% test accuracy.
- Developed student-performance classification pipelines with train-only preprocessing, leakage-column exclusion and a stratified evaluation split; reproduced 94.75% Logistic Regression accuracy on the supplied coursework dataset.
- Applied K-Means/PCA for segmentation and tabular Q-learning for deterministic/stochastic Four Rooms navigation. Coursework results are not externally validated deployment claims.

Foundry AI Infographic Dashboard | Foundry, Ontology, AIP Logic, Gemini | 2026

- Designed data transformations and Ontology-backed analytical workflows for enterprise KPI exploration.
- Implemented natural-language interactions and Gemini-generated infographic summaries with dashboard, query and generation-history interfaces.

TuToR Marketplace | React Native, Expo, Firebase | Collaborative UCT capstone, 2024

- Contributed to mobile navigation, authentication, chat, meeting and profile workflows; Git history records 40 commits under the existing Azhar identity across shared application screens.
- Worked within a collaborative student/tutor application using Firebase authentication and persistence across shared mobile workflows.

Embedded and Systems Coursework | C, STM32, Java | UCT, 2023 / 2024

- Worked on collaborative STM32 practicals using timer-triggered DMA, waveform lookup tables, PWM and ADC; original hardware/toolchain is required.
- Implemented and analysed Java FCFS/SJF scheduling and address translation on teaching scaffolds; explored thread coordination, queues and timing tradeoffs.
- Built C++ connected-component processing with thresholding, queue-based flood fill and component-size filtering; cleanup verification passed 19 assertions across two test cases.

EDUCATION

BSc Honours Security & Network Engineering | Eduvos

Current, part-time online study.

BSc Computer Science & Computer Engineering | University of Cape Town

Graduated 2025. Software, algorithms, databases, networks, embedded systems and AI/ML. Earlier Mechatronics study before transferring.

CERTIFICATIONS AND TRAINING

- **Ontology University - Junior Forward Deployed Engineer:** Certificate of Achievement, June 2026.
- **SAP Certified Associate - Business Process Integration with SAP S/4HANA:** earned January 2025; supplied credential expired January 2026.
- **Microsoft AZ-500, SC-200 and SC-5002:** course/training records, not verified professional exam certifications.
- **SAP ABAP on SAP BTP / Basic ERP Processes:** learning records and training, 2024-2025.